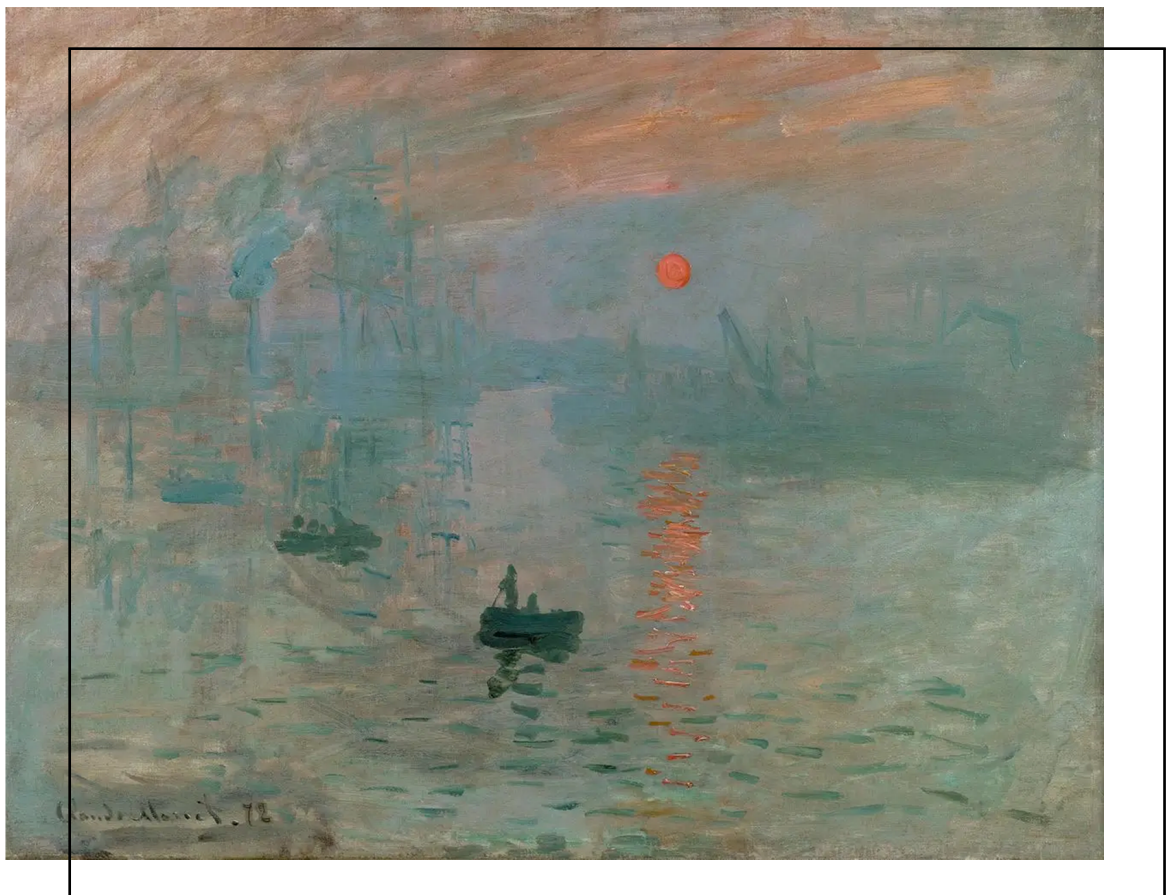


01

Lista de Exercícios

Relatividade Geral - A Teoria da Gravitação de Einstein

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Q. 01

Gravitational Field: a sphere, pt. 1

Find the gravitational field inside and outside a sphere of radius R , which carries a uniform volume mass density ρ .



Q. 02

Gravitational Field: a spherical shell, pt. 1

Find the gravitational field inside and outside a spherical shell of radius R , which carries a uniform surface mass density σ .



Q. 03

Gravitational potential: a straight wire

Find the gravitational potential a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compute the gradient of your potential and find the corresponding gravitational field.



Q. 04 Gravitational potential: a hollow sphere

The gravitational potential $U(\theta)$ is specified on the surface of a hollow sphere of radius R . Find the potential inside the sphere.



Q. 05

Gravitational potential: a sphere, pt. 2

The potential at the surface of a sphere of radius R is given by $U(\theta) = k \cos(3\theta)$, where k is a constant. Find the potential inside and outside the sphere, as well as the surface mass density $\sigma(\theta)$ on the sphere (assume there is no mass inside or outside the sphere).



Q. 06 Gravitational potential: a spherical shell, pt. 2

Derive the exact gravitational potential for a spherical shell of radius R which carries a surface mass density $\sigma(\theta) = k \cos(\theta)$, where k is a constant. Calculate also the quadrupole moment of this mass distribution.
